

Acute Otitis Media - Summary

Key Symptoms: Inflammation of inner ear with signs of infection: ear pain, discharge, fever, etc.

Most common causing-pathogens: Majority are bacterial caused:

- *S. pneumoniae*: occurrence rate of 25-50% with roughly 20% of cases self-resolve
- *H. influenzae*: occurrence rate of 15-30% with roughly 50% of cases self-resolve
- *M. catarrhalis*: occurrence rate of 3-20% with roughly 75% of cases self-resolve

Non-Pharmacological Treatment:

- Consider watchful waiting in patients in the first 24 - 48 hours in children ≥ 6 months of age with the following: mild illness, Uncomplicated AOM, temperature $< 39^{\circ}\text{C}$, no serious comorbidities (e.g. immunodeficiency), guardian can recognize when illness is worsening and return for follow-up
- **Acetaminophen**: used for analgesia and antipyretic effects
- **Ibuprofen**: used for analgesia and antipyretic effects

Pharmacological Treatment:

- **Key Points:**
 - ≥ 2 years (uncomplicated): 5 days Tx
 - < 2 years and/ or relapse, failure/ perforated tympanic membrane: 10 days Tx
- **First-line therapy:**
 - **Standard-Dose (SD) Amoxicillin**: 1st line, no daycare exposure, no abx use in past 90 days.
 - **High-Dose (HD) Amoxicillin**: used to overcome *S. pneumoniae* resistance (use if recent abx exposure, attends daycare or have siblings in daycare)
- **First-line therapy in patients with non-life-threatening penicillin allergy:**
 - **Cefuroxime axetil or ceftriaxone**
- Initial treatment failure (no improvement after 2-3 days):
 - **Amoxicillin-clavulanate**: Preferred suspension in Canada is the 7:1 (amoxicillin: clavulanate) formulation
 - **Ceftriaxone**: Reserved for patients who cannot tolerate oral antibiotics or have failed amoxicillin-clavulanate

Monitoring Parameters: Parameters similar to watchful waiting, see slide 18 in AOM lecture

Additional Key Points:

- Clavulanic acid increases risk of diarrhea
- Ceftriaxone follows a 3-day regimen rather than 5-10 days